

PROJECT PROPOSAL

Smart Recipe Planner & Pantry Manager

Cook from what you already have — less waste, less guesswork.

Item	Detail
Team	Apex
Project title	Smart Recipe Planner & Pantry Manager (What2Cook)
Document	Project Proposal (DISCOVER phase deliverable)
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1. Executive Summary

The Smart Recipe Planner & Pantry Manager is a mobile-first web application that helps home cooks decide what to cook using the ingredients they already own. Users log their pantry items with quantities and expiry dates, and the system generates personalised recipes and weekly meal plans around the food that is about to expire, while respecting their dietary preferences. Planned meals are converted automatically into a grocery list that contains only the items actually missing from the pantry.

Households throw away a large share of the food they buy — not deliberately, but because meal decisions are made separately from the food already in the kitchen. Recipe websites are organised around dishes rather than around available ingredients, and meal plans, saved recipes and shopping lists live in different places. The result is predictable: people overbuy, cook the same few meals, and discard edible food every week.

This proposal describes a single, tightly scoped workflow — pantry in, meal plan and grocery list out — delivered over a 14-week programme by a five-member team. The system is designed under Thai law from the first sprint: dietary and allergy data is treated as sensitive personal data under the PDPA, every content action is logged under the Computer Crime Act §26, and every consent and AI generation is stored as a retrievable, reproducible record under the Electronic Transactions Act.

Success is measured, not asserted. In a before/after pilot with real users, the project targets a median reduction of at least 30% in self-reported food thrown away per household per week, and a drop in median weekly meal-planning time to under 15 minutes, measured over at least two weeks per user.

2. Problem Statement

Users of the current, informal way of planning meals face three connected obstacles. These were confirmed in user interviews and are referenced throughout this proposal as pains P1–P3.

ID	Pain	Description
P1	Invisible inventory	People do not know what they already have or when it expires. Ingredients are pushed to the back of the fridge and discovered only after they have spoiled, so food is thrown away and re-bought.
P2	Decision fatigue and unusable recipes	Recipe sites are organised around dishes, not around available ingredients. Users must read a recipe first and then check whether they can make it — and most recipes demand a shopping trip for one or two missing items, so people default to takeout.
P3	Disconnected planning and shopping	Meal plans, saved recipes and grocery lists live in notes apps, screenshots, bookmarks and memory. Shopping lists are written from scratch and duplicate items already in the pantry.

The consequence is a weekly loss of both money and time: users overbuy, cook the same few meals repeatedly, and throw away edible food.

3. Project Goals and Objectives

Goal. Help users cook from what they already have, so that less food is wasted and weekly planning takes less effort.

3.1 Objectives

- **Make the pantry visible** — one place where users see what they have, how much, and what expires soonest.
- **Cook from what is on hand** — use AI to generate recipes built around the user's current pantry, prioritising items close to expiry.
- **Respect dietary needs** — apply saved dietary filters (vegetarian, gluten-free, low-carb, allergies) to every generated or suggested recipe, with zero tolerance for violations.
- **Streamline grocery shopping** — convert planned meals into an accurate shopping list that excludes anything already in the pantry.
- **Reduce food waste measurably** — achieve a documented before/after reduction in discarded food and in weekly planning time with real users.
- **Comply by design** — satisfy PDPA, Computer Crime Act §26, and Electronic Transactions Act obligations as build requirements, not as a later audit.

3.2 Success criteria

ID	Criterion	Target
NFR1	Food waste and planning time (the project metric)	Waste down $\geq 30\%$; planning time under 15 min/week; ≥ 2 weeks per user
NFR2	Pantry entry friction	≤ 3 fields, < 15 s per item; 20 items in < 6 min
NFR3	Recipe generation latency	< 10 s at p90; progress shown within 1 s
NFR4	AI fidelity — pantry	$\geq 90\%$ pantry-only recipes; $\geq 80\%$ use an expiring item
NFR5	AI fidelity — dietary	100% — any violation is a release blocker
NFR6	Grocery list accuracy	0 already-stocked items across ≥ 20 test plans
NFR7	Availability	$\geq 99.0\%$ per calendar month; plain-language failure notice

ID	Criterion	Target
NFR8	Cost control	30 generations per user per month on the free tier, enforced server-side

4. Scope

4.1 In scope — the core workflow

The system delivers one workflow end to end: pantry in → AI recipe → meal calendar → grocery list out. Eight core features support it.

#	Feature	Description	Priority
1	Pantry Tracker	Log what food you have, how much, and when it expires; sorted expiry-first.	Core
2	Calendar Meal Planner	Assign a saved or generated recipe to each day of the week.	Core
3	AI Recipe Generator	AI creates a recipe from pantry contents, respecting dietary filters.	Core
4	Auto Grocery List	Turn planned meals into a shopping list, minus what is already stocked.	Core
5	Recipe Saving	Save AI-generated and manually entered recipes for reuse.	Core
6	Dietary Filters	Filter recipes by vegetarian, gluten-free, low-carb and allergy needs.	Core
7	Recipe Import (Manual)	Write a recipe by hand into the app's form.	Core
8	Search by Ingredient	Find saved recipes using ingredients already on hand.	Core

4.2 Out of scope for this phase

- Integration with a licensed Certification Authority, and any "certified" or "nutritionist-approved" claim in the product (LR9).
- Barcode or receipt scanning for pantry entry, nutritional analysis, and cost tracking.
- Grocery retailer / delivery integrations and multi-household shared pantries.
- Native iOS and Android applications — the deliverable is a mobile-first web app.
- A self-hosted or fine-tuned recipe model (see the architecture trade-off in section 6.2).

5. Feature Breakdown and Acceptance Criteria

Requirements state what a user wants; features state what gets built. Each feature below is testable — a QA tester can pass or fail every criterion.

ID	Feature	Key acceptance criteria	Traces
FE1	Pantry Management	Item saved in ≤ 3 fields; expiry-first ordering; "expiring soon" marking within 3 days; edit and delete; 200 items render in < 3 s.	F1, F2, F12; NFR2, NFR9

ID	Feature	Key acceptance criteria	Traces
FE2	Dietary Profile & Consent Gate	No dietary flag set without an explicit opt-in record; Terms and AI & Food Safety Disclaimer accepted before first generation; AI-transfer consent separate from Terms; each acceptance and withdrawal stored with user id, timestamp and version.	F4, F8; LR1, LR2, LR7
FE3	AI Recipe Generation	Pantry-only ingredients plus declared staples in $\geq 90\%$ of a 50-generation test set; $\geq 80\%$ use an expiring item; 100% honour dietary filters; outbound payload carries ingredients and dietary flags only; every generation stored for reproducibility; 30/month cap.	F3; NFR3–NFR5, NFR8; LR2, LR8
FE4	Recipe Library	Save and reopen an AI recipe without regenerating; manual recipe form; search by ingredient; create/edit/delete logged separately from content; originals kept ≥ 90 days.	F7, F10, F11; LR4, LR5
FE5	Weekly Meal Planner	Assign a recipe to a day; plan renders in < 3 s; marking a meal "cooked" deducts pantry quantities; plan changes allowed; "meal plan saved" stored as a retrievable record.	F5, F9, F12; LR8
FE6	Auto Grocery List	List derived from the planned week minus current stock; 0 already-stocked items across ≥ 20 test plans; "grocery list generated" stored as a record.	F6; NFR6, LR8
FE7	Compliance & Audit Layer	Cross-cutting: every create/edit/delete writes account id, IP and timestamp to a separate log store retained ≥ 90 days; extended-retention flag up to 1 year; account deletion removes user content but not logs; login stores a token only.	LR3–LR6, LR9; NFR10

Build order for the core workflow: FE1 → FE2 → FE3 → FE4 (save) → FE5 (assign) → FE6, with FE7 running behind all of them.

6. Proposed Solution and Architecture

6.1 System architecture

The application is a mobile-first web client over a service-oriented backend and four separate data stores. Compliance is placed in the call graph rather than in policy documents.

Layer / component	Responsibility
Client (web app)	Screens S1–S6: pantry, add item, consent, recipe result, weekly plan, grocery list.
Pantry Service	Pantry CRUD, expiry ordering, deduct-on-cooked (F1, F2, F9).
Recipe Service	Recipe library, manual entry, ingredient search (F7, F10, F11).
Meal Plan Service	Weekly calendar assignment and changes (F5).
Grocery List Service	Derives the list at request time: planned meals minus current stock (F6).
Consent Service	Dietary opt-in, Terms acceptance, AI-transfer consent and withdrawal (LR1, LR2, LR7).
AI Gateway	The single outbound chokepoint: strips identity, enforces the quota, writes the generation record (LR2, LR8, NFR8).
Audit & Log Service	Writes account id, IP and timestamp for every create/edit/delete (LR4–LR6).
App DB	Pantry, recipes, plans, and the dietary profile (encrypted at rest).
Consent store	Append-only acceptance and consent records.
Log store	Access and action logs, retained ≥ 90 days by explicit configuration.
Generation store	AI output text, model version, pantry snapshot, filters, timestamp.

6.2 Key design decisions

- **The AI Gateway is its own component.** One chokepoint where identity is stripped, quota is enforced and the generation record is written — compliance in one testable place instead of scattered through the app.
- **Four stores, not one.** Logs must survive content deletion and consent must be append-only; different lifecycles cannot share a table.
- **The Consent Service gates the gateway.** An ungated path to the AI service does not exist in the call graph.
- **The grocery list derives, never stores.** It is computed from plan minus pantry at request time, so it cannot drift out of date.
- **The dietary profile is encrypted at rest.** It is sensitive personal data under the PDPA.

6.3 Stated trade-off: third-party AI service instead of a self-hosted model

We gain: a working recipe generator in month 2 with no ML work, no GPU cost, and output quality good enough to test the real hypothesis — that cooking from the pantry reduces waste. A five-person team with a three-week build window cannot ship a self-hosted model and still deliver the workflow.

We pay: per-call cost and rate limits (NFR8 caps free-tier users at 30 generations per month), an availability dependency we do not control (NFR7 sets a 99% target and requires a plain-language error, never a raw one), and a PDPA cross-border transfer that requires disclosure, consent and data minimisation (LR2).

Revisit when: generation cost per active user exceeds the free tier at scale, or an allergen-safety failure traces to model behaviour we cannot constrain by prompt.

7. User Journey

Persona — Nan. Works full time, shops once a week, throws out produce most weeks, and orders takeout when she cannot decide what to cook (P1, P2, P3).

#	Screen	Nan does	System does
J1	Sign-up / Terms	Creates an account and accepts the Terms and the AI & Food Safety Disclaimer.	Stores acceptance: user id, timestamp, version.
J2	Dietary profile	Opts in to gluten-free and a peanut allergy.	Stores it as sensitive data under a separate explicit opt-in; never set by default.
J3	Add pantry item	Types "spinach, 1 bag, expires Sep 4" — three fields, about 12 seconds.	Saves the item; ~15 items logged in under 6 minutes.
J4	Pantry list	Opens the app on Wednesday evening.	Shows the pantry expiry-first; spinach and eggs flagged "expiring soon".
J5	Consent	Taps "Cook something with this" for the first time.	Asks consent to send ingredient names and dietary flags, and explains what is not sent.
J6	Recipe result	Waits about 6 seconds.	Returns a spinach-and-egg recipe using pantry items and staples only, with zero gluten and zero peanuts; stores the generation record.
J7	Recipe result	Taps "Save".	Saves to the library; logs account id, IP and timestamp separately from the content.
J8	Weekly calendar	Drags the recipe onto Thursday and adds two more meals.	Stores the plan and returns a retrievable "meal plan saved" record.
J9	Grocery list	Taps "Build my shopping list".	Returns the 6 missing items instead of all 20, because pantry stock is subtracted.
J10	Cooked	Marks Thursday's meal "cooked".	Deducts the used quantities so the pantry stays accurate for next week.

7.1 Alternate and failure paths

- AI service down or rate-limited: a plain-language notice, never a raw error (NFR7).
- Free-tier cap reached: generation is blocked server-side, the cap is explained, and saved-recipe search is offered (NFR8).

- Pantry empty: the user is prompted to add items rather than generating from nothing.
- Consent withdrawn: generation is disabled and the withdrawal is stored the same way the consent was (LR7).
- Account deleted: pantry, plan, recipes and dietary profile are deleted; access logs are retained ≥ 90 days as the law requires.
- A generated recipe contains an allergen: release blocker — NFR5 is 100%, not a target.

8. Prototype

A low-fidelity wireframe set fixes what appears on each screen and why; colour, type and spacing are the Designer's decision during BUILD. No production code is written in the DISCOVER phase.

Screen	Name	Purpose	Traces
S1	Pantry (home)	Expiry-first list with "expiring soon" grouping and the primary "Cook something with this" action — the pantry is the entry point, not a sub-page.	F1, F2
S2	Add item	Three fields and "Save & add next", keeping focus in the form so 20 items take under 6 minutes. This screen carries the adoption risk.	F1, NFR2
S3	Consent	Shown once before the first generation: exactly what is sent (ingredient names, dietary flags) and what is not (name, email, household).	F8, LR2, LR7
S4	Recipe result	Generated recipe with the pantry items it used, the expiring items it prioritised, and a Save action.	F3, F7
S5	Weekly plan	Seven-day calendar with one assignable meal slot per day and a "cooked" action.	F5, F9
S6	Grocery list	Only the missing items, grouped for shopping, derived from the plan minus stock.	F6

9. Legal and Compliance Requirements

Three Thai laws govern this system. Each obligation below is carried into the backlog as a Must item, so compliance is verified by acceptance testing rather than by review at the end.

9.1 Personal Data Protection Act (PDPA)

- Dietary filters that can reveal health or religion (gluten-free/coeliac, diabetic, halal, allergy lists) are treated as sensitive personal data and collected under explicit, separate opt-in consent — never as a silent default (LR1).
- Sending pantry items and dietary flags to a third-party AI service is disclosed, consented to before the first generation, and minimised to ingredient names plus dietary flags — never the account email, name, or household identifiers (LR2).
- Pantry data is used only for recipe generation, meal planning and the grocery list — not for advertising, resale or profiling (LR3).
- Account deletion removes pantry items, meal plans, saved recipes and the dietary profile, and they are not retained in backups beyond the stated window; access logs are the documented exception (LR3).
- Social or email login stores a verification token only, never a provider credential (LR3).
- A shared or exported grocery list never carries another person's dietary or health data without that person's own consent.

9.2 Computer Crime Act §26

- Creating content (a manual recipe, a saved AI recipe, a shared plan) logs the account ID, IP address and timestamp separately from the content itself (LR4).
- Edit and delete actions log actor identity, IP and timestamp, kept ≥ 90 days even after the item is deleted (LR5).
- A record can be flagged for extended retention up to one year on a competent authority's request, without altering its content (LR6).
- A taken-down recipe is still retained internally, with its log, for the required period.
- Log retention on cloud infrastructure is configured explicitly to ≥ 90 days, since cloud defaults are usually shorter (NFR10).

9.3 Electronic Transactions Act §9 / 26 / 28

- Accepting the Terms of Use and the AI & Food Safety Disclaimer records user ID, timestamp and terms version as a retrievable acceptance record (LR7).
- Consent for sensitive dietary data and for AI transfer — and its withdrawal — is stored as a retrievable, timestamped, versioned record (LR7).
- Every AI-generated recipe stores its exact text, the model version, the pantry snapshot and the dietary filters used, so the output can be reproduced in a later dispute (LR8).
- Confirmations such as "meal plan saved" and "grocery list generated" are stored as retrievable records, not transient UI notices (LR8).

- No "certified", "nutritionist-approved" or CA-backed signature language appears anywhere, because no licensed Certification Authority or certified professional is involved (LR9).

10. Project Team and Responsibilities

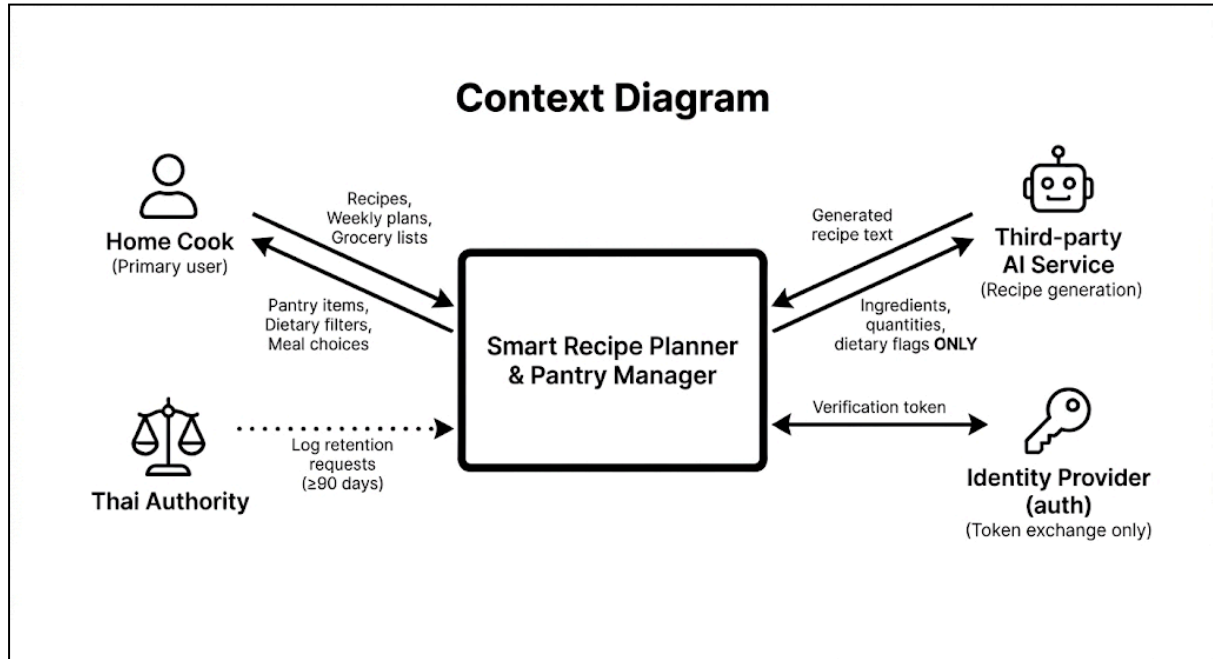
Team Apex is a five-member team. Each member owns one area end to end and is accountable for the deliverables listed against that role.

Student ID	Name	Role	Responsibilities
6631503056	Chell Hmue May	Project Owner	Owens the project vision; defines requirements; manages scope, priorities and timeline.
6631503057	Eaindray Su Pan	AI Lead	Owens the team's AI tooling; builds agents.md, skills.md, the spec and the backlog; handles AI disclosure and governance.
6631503063	Kyu Kyu Thin	QA / Tester	Verifies specs against the backlog; checks traceability and legal requirements; tests that acceptance criteria are met.
6631503120	Hsu Myat Thwe	Tech Lead	Oversees the GitHub repository, code standards and technical decisions.
6631503096	Thwin Khant Nyar Zaw	Designer	Creates user-friendly, visually appealing and intuitive digital experiences; owns the wireframes and the visual prototype.

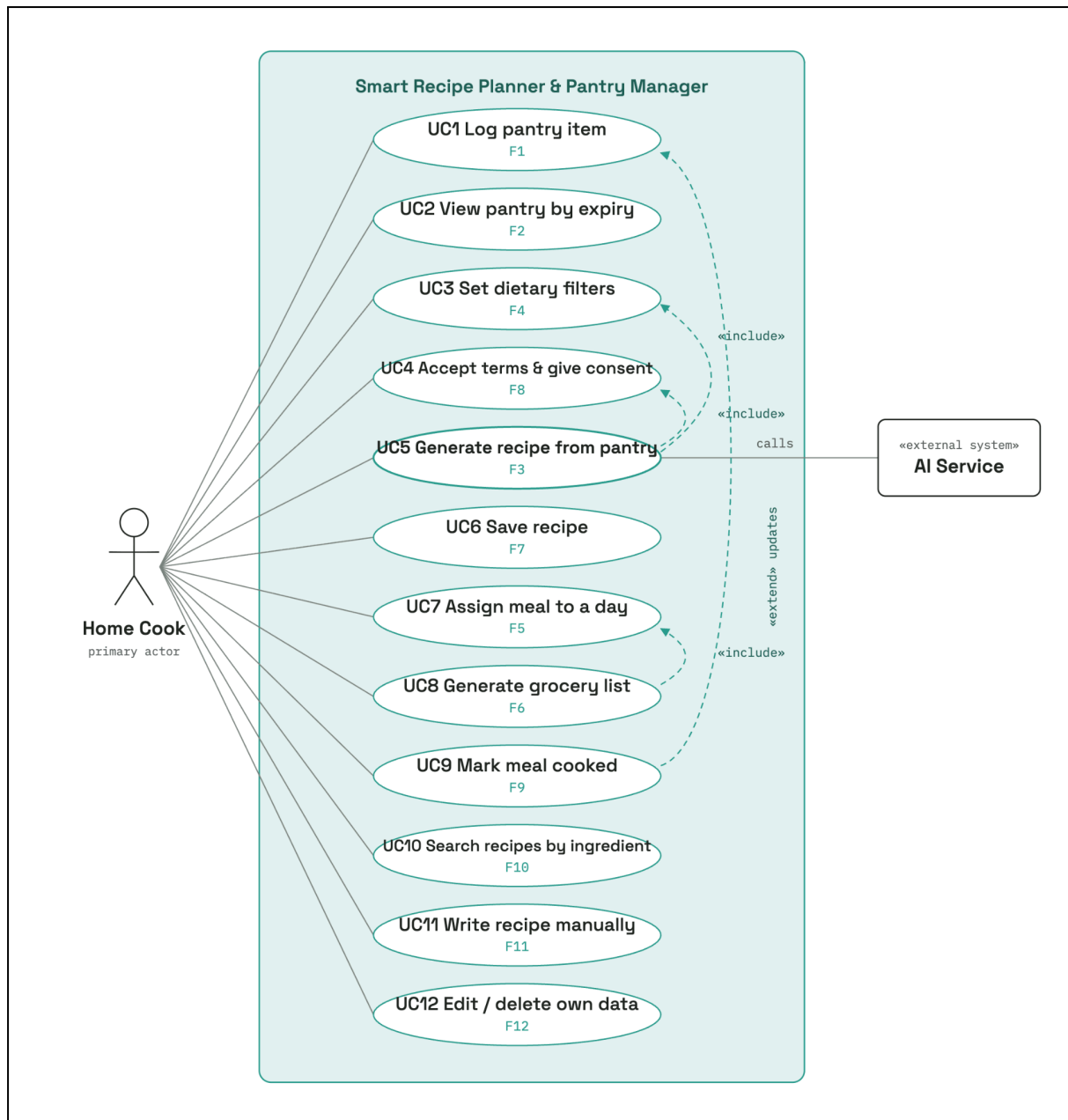
10.1 Other stakeholders

Stakeholder	Interest in the project
Target users — home cooks and students	Track their pantry, plan weekly meals, and shop from the generated list with less waste.
Household members	Shop from a list someone else made; their dietary data must not be exposed without their own consent.
Course instructors / assessors	Evidence of traceability from interview pain to requirement, feature, acceptance criterion and test.
Third-party AI service provider	Usage volume, cost and rate limits; the subject of the PDPA cross-border transfer disclosure.

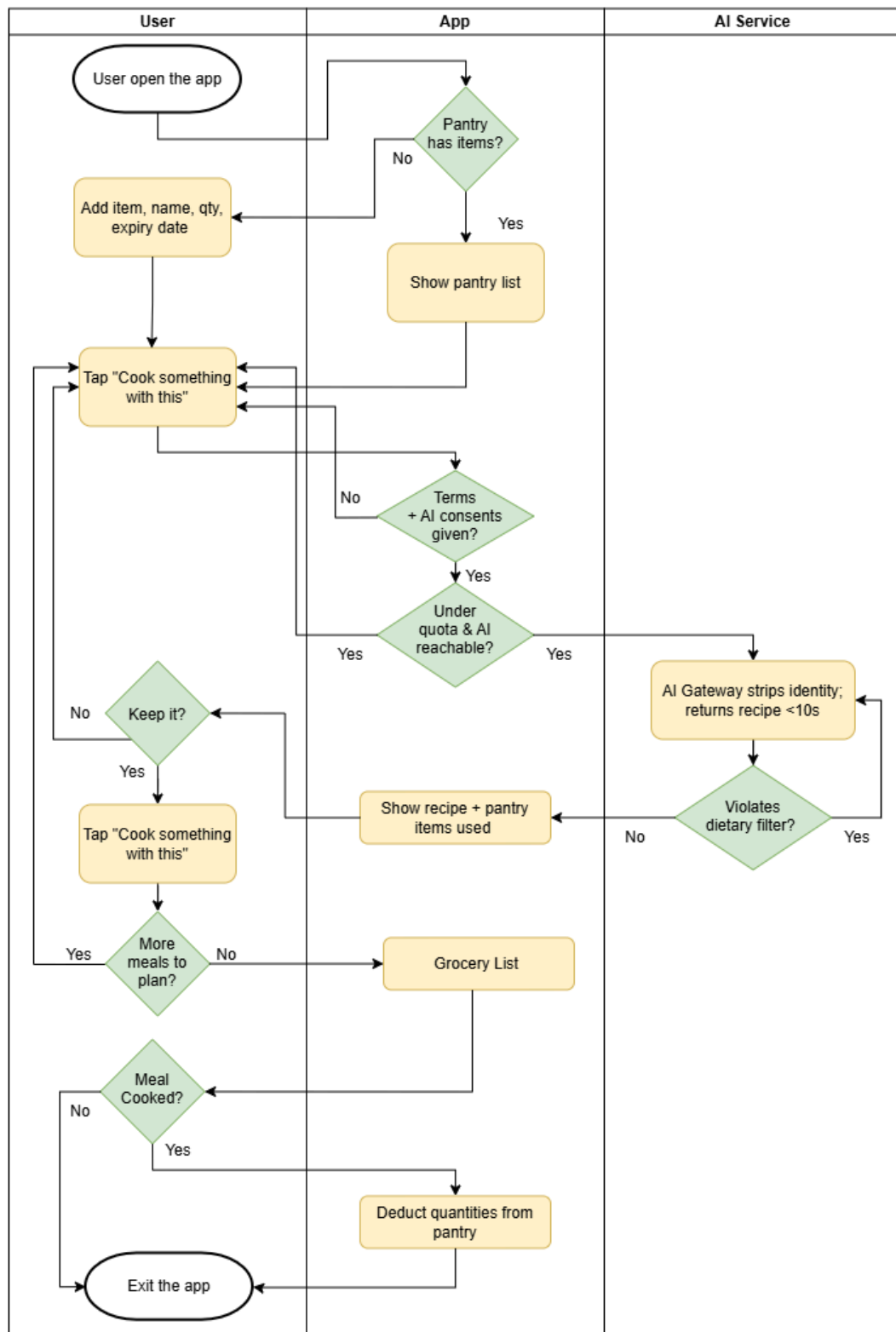
11. Diagrams



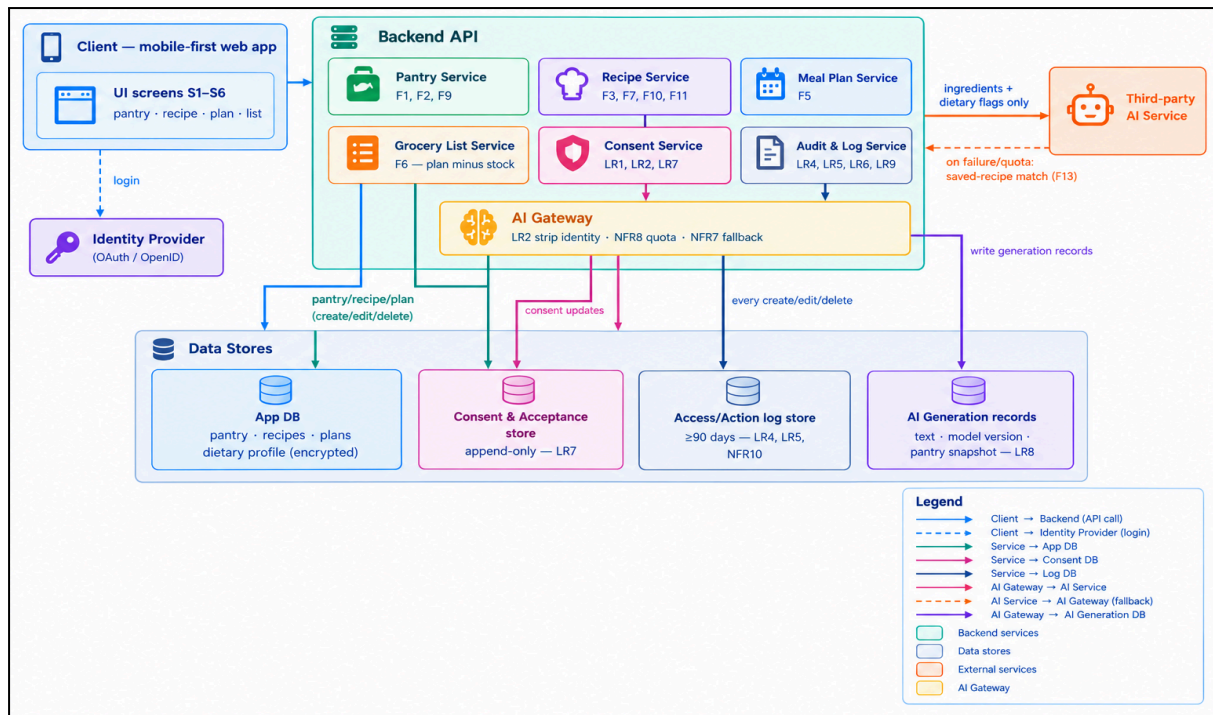
11.1. Context Diagram



11.2. Use-case Diagram



11.3. Activity Diagram



11.4. Architecture Diagram

12. Timeline and Milestones

Phase	Weeks	Milestone and deliverables
Discover	W1–W5	Project charter; requirement spec and backlog; four diagrams and a prototype sketch; User Validation Gate (pass/fail).
Build	W6–W8	Scope locked and Sprint 0; Alpha Demo — a user adds pantry items, generates an AI recipe from them, assigns it to a day, and receives a grocery list end to end.
Test	W9–W11	Beta Review — before/after metric on food waste and planning time, plus at least three feedback-driven fixes (Impact Report).
Deliver	W12–W14	Real-user sign-off; Final Showcase (10-minute pitch); portfolio pack — case study, demo video, evidence log, impact metrics and role statements.

12.1 Deliverables checklist

- Project charter, requirement specification and prioritised backlog (Discover).
- Context, use-case, architecture and activity diagrams, plus the low-fidelity prototype (Discover).
- Working application covering the core workflow FE1–FE6 with FE7 behind it (Build).
- Test evidence: the 50-generation AI test set, the 20-plan grocery-list check, and the compliance log verification (Test).
- Impact report with before/after metrics, and the final portfolio pack (Deliver).

13. Risks, Constraints and Mitigations

Risk / constraint	Why it matters	Mitigation
AI output quality	Hallucinated quantities or an ignored dietary restriction would break user trust and could be unsafe.	NFR4 and NFR5 are measured on a 50-generation test set; a dietary violation blocks release; every generation is stored so any failure is reproducible.
Manual entry friction	Typing every item with quantity and expiry is the biggest adoption barrier — users abandon the app in week 1.	Three fields only, a 15-second target per item, and "Save & add next" to keep focus in the form (NFR2).
Pantry data accuracy	If stock is not deducted after cooking, recipes and grocery lists become wrong.	Deduct-on-cooked (F9) makes accuracy a by-product of normal use rather than a separate chore.
AI cost and availability	Generation depends on a third-party service with usage costs and rate limits.	Server-side cap of 30 generations per user per month (NFR8); graceful degradation to saved-recipe search with a plain-language notice (NFR7).
Measurement difficulty	Food waste reduction is self-reported, so the headline metric could be challenged.	NFR1 defines the method up front: median per household per week, before/after, ≥ 2 weeks per user, with planning time as a second measure.
Consent prompt friction	A legal wall before the first generation could scare users off at the key moment.	Show exactly what is sent (ingredients and dietary flags) and what is not, instead of a legal blob (LR2).
Compliance scope creep	Logging, consent records and retention touch every feature and can be deferred until it is too late.	FE7 is a cross-cutting Must feature built alongside FE1–FE6, with its own acceptance criteria.
Team capacity	Five members, a three-week build window, and a fixed course schedule.	Scope locked at Sprint 0; the Build commitment is the single core workflow, and everything else waits.

14. Conclusion

The Smart Recipe Planner & Pantry Manager addresses a common and measurable household problem with a single, coherent workflow rather than a broad feature set. By connecting the pantry, the meal calendar and the shopping list in one place, and by generating recipes from what is actually on hand and closest to expiry, the system removes the daily "what should I cook?" decision and the weekly cycle of overbuying and discarding.

The scope is deliberately narrow and every requirement is traceable from an interview pain or a legal obligation through to a testable acceptance criterion. Thai PDPA, Computer Crime Act and Electronic Transactions Act obligations are treated as build requirements from the first sprint. Success will be judged on a real before/after measurement with real users, not on the feature list.

15. Authorisation

The undersigned team members approve this proposal and commit to the scope, timeline and responsibilities described above.

Student ID	Name	Role	Signature / Date
6631503056	Chell Hmue May	Project Owner	6 Sep 2026
6631503057	Eaindray Su Pan	AI Lead	6 Sep 2026
6631503063	Kyu Kyu Thin	QA / Tester	6 Sep 2026
6631503120	Hsu Myat Thwe	Tech Lead	6 Sep 2026
6631503096	Thwin Khant Nyar Zaw	Designer	6 Sep 2026